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TUM SCHOOL OF SOCIAL SCIENCES & TECHNOLOGY · SOT86140

Quantum Hackathon **Syllabus**

Applications of Quantum Computing in Life Sciences, Finance, Mobility & GovTech — a guided, interdisciplinary project module.

Semester Summer 2026 / 27

Pre-learning 05–07 Oct

Sprint 15–16 Oct

Final pitch 30 Oct

Platform Moodle · TUMonline

Subject to change — version 01 October 2026

Visual design generated with AI · all course content authored by the teaching team

SECTION 01

Course overview & learning objectives

This hackathon runs as a guided project — you won't be thrown in at the deep end, and you won't be handed a recipe to follow either.

We give you the foundations: a shared grounding in quantum concepts and the software tools you'll need, clearly bounded challenge spaces, prepared data and technical resources, and mentors to turn to along the way.

Inside those boundaries, the real work is yours. You'll decide how to solve your specific challenge, choose a method and justify why it fits, build a solution someone else could reproduce, and judge honestly how far it goes — and where it falls short. The project moves the way real problem-solving does: understanding and framing the challenge first, then prototyping, testing, and refining.

You'll look at every challenge from four angles at once — the domain it comes from, the classical computational approach, the quantum perspective, and the wider societal context it sits in. One principle runs through all of it: **technical complexity on its own does not make a strong project.** What makes a project strong is a credible fit between the problem, the method you chose, the evidence you produced, and the setting where your solution would actually be used.

By the end, you'll be able to

- Turn a messy real-world challenge into a clear computational problem, workflow or project plan
- Select and justify an appropriate quantum or hybrid approach
- Build and evaluate reproducible computational workflows
- Assess technical feasibility under realistic constraints
- Reflect critically on the societal, political, and ethical implications of quantum technologies
- Communicate your solution — the evidence, the trade-offs, the limitations — to technical and non-technical audiences alike

You'll work in interdisciplinary teams, and you'll present your project at the end of the module — showing not just what the team built, but your own command of it and your part in getting there.

SECTION 02

Administrative details

Course format

The module is offered as a block seminar structured in three parts:

PHASE I
05 – 07 Oct

Pre-learning

Conceptual and practical foundations through introductory lectures and guided exercises. No prior experience in quantum computing required.

PHASE II
15 – 16 Oct

Sprint

A two-day intensive project sprint where students engage with a state-of-the-art challenge and develop an initial solution concept and project plan.

PHASE III
after the sprint

Extended project

Teams continue developing their projects in small interdisciplinary groups, with a one-time mentoring and feedback session before the final presentation.

TEACHING MATERIALS	Seminar, tutorials, papers, use cases and ad-hoc materials
COMMUNICATION	Moodle is the central platform for announcements, materials, deadlines and organisational information. Check it regularly for updates and course-related communication.

Dates & time

PRE-LEARNING PHASE	05.10 · 06.10 · 07.10 — 15:30 – 19:00
SPRINT PHASE	15.10 – 16.10 — 10:00 – 19:00
FINAL PRESENTATION	30.10 — 15:00 – 16:30 (internal) & 17:00 – 20:00 (external, with guests)
LOCATION	Please check TUMonline.

Teaching team

INSTRUCTORS	Fabienne Marco · fabienne.marco@tum.de Eden Castaneira Rios · eden.castaneira@tum.de Gabriela Boechat · Gabriela.Boechat@tum.de
TEACHING ASSISTANTS	Ana-Maria Cupes · anamaria.cupes@tum.de Benedikt Wimmer · ben.wimmer@tum.de

SECTION 03

Examination

The final presentation session takes place on **30.10**. Grading is split across two deliverables:

40%

Intermediate · sprint outcome

A 15-minute presentation per team plus a PowerPoint document, an initial solution architecture and project plan developed during the two-day sprint. This intermediate outcome is the foundation for the project phase, during which teams refine, validate and operationalize their approach.

60%

Final · project pitch

A 20-minute presentation and PowerPoint document — a final project pitch demonstrating the developed solution and its implementation strategy; the code via GitHub is also graded. Assessment during the pitch is individual, considering performance and the capacity to answer project-related questions from each contribution and the broader implications of the approach.

The final presentation brings together the problem, methodology, implementation, results, feasibility, limitations and broader implications of the project as one coherent story. Structured as a classical pitch, teams may choose the format that best supports their project — a slideshow, live demo, or other presentation format.

> Teams should demonstrate their code or a reproducible workflow and explain how they responded to the feedback and recommendations from the **Policy Practice Team**, including how these insights refined or improved the outcome. Each student presents and answers questions about both their individual contribution and the overall project logic.

SECTION 04

Syllabus & preparation

Team matching · challenges & policy advice

Team matching

Teams are pre-formed by the Hackathon Team based on the survey on your pre-knowledge. If you already filled in the survey, you can find the team matching on the Moodle page.

Challenges & policy advice

The hackathon features four challenge tracks. Each team works on a challenge within its assigned domain, exploring potential applications of quantum computing and assessing their feasibility, limitations and broader implications. Tracks are assigned in early October, so teams can focus on their topics from the start of the pre-learning phase.

- **Mobility**
- **Finance**
- **Life Sciences**
- **GovTech**

In addition to the technical work, teams receive policy advice from the Policy Practice Team — students from the Global Technology Policy Practice course — matched with hackathon teams by thematic track. They provide recommendations on relevant policy considerations, potential risks and implementation challenges. Teams are expected to engage with this advice throughout the project and reflect on how it informs their solutions, including the measures needed to address identified risks and support potential implementation.

SECTION 05

Part I — Pre-learning phase

05 – 07 October · 15:30 – 19:00 daily

DAY 1 — Foundations & framing

05 October

TIME	CONTENT
15:30 – 16:00	Introduction to the Hackathon — how the format works, expected workload, goal and everything around the organisation. *Eden Castaneira, Fabienne Marco & Gabriela Boechat*
16:00 – 16:30	Quantum Hype — where we stand with the development of quantum technologies across Finance, Life Sciences, Mobility and GovTech. *Fabienne Marco*
16:30 – 17:30	Introduction to Quantum Mechanics. *Ginevra Fulco*
17:30 – 18:00	*Break*
18:00 – 19:00	Translating a problem into technical terms — modelling a decision. Binary variables, objective versus constraints, hard versus soft, and penalty terms as the place where a non-negotiable requirement turns into a number a better objective can outweigh. *Eden Castaneira*

DAY 2 — QUBO, Ising & QAOA

06 October

TIME	CONTENT
15:30 – 16:00	QUBO and Ising: translation from the Day 1 decision model, penalty weights, slack variables, unbalanced penalisation. Baselines and exact methods — greedy and top-k with one case where each goes wrong; CP-SAT or HiGHS in twenty lines; the exact solution as verification, not competition.
16:00 – 16:45	*Break*
17:15 – 18:00	QAOA: cost and mixer Hamiltonians, parameters, sampling, reading a bitstring distribution, and why a sampler that returns many answers can be a feature. Ten minutes on quantum-inspired methods (simulated annealing, simulated bifurcation) as legitimate entries.
18:00 – 19:00	QUBO hands-on session.

DAY 3

Responsibility & domain sessions

07 October

TIME	CONTENT
15:30 – 16:00	Responsibility (all tracks together). What the model decides on someone's behalf: who is stranded, what is discarded, which limit got traded. The error that produces evidence versus the one that leaves no trace. Explaining a result to the person who has to act on it.
16:00 – 17:30	Domain-specific knowledge sessions — parallel by track (see below).
17:45 – 19:00	Plenary session — challenge walkthrough per track: the brief, the starting package, the methods menu and one worked tiny instance, so every team leaves knowing its challenge, its data and its first step.

Domain-specific sessions · 16:00 – 17:30 (parallel by track)

GovTech

Gabriela Boechat

Introduces GovTech and the role of state capacity in enabling governments to work with emerging technologies. Examines regulatory sandboxes and the emerging framework for experimentation in Germany, including the institutional and legal challenges of putting experimentation into practice.

Mobility

Helena Gartmeier

Introduces public transport planning as an allocation problem under changing demand, limited capacity and operational constraints. Examines how routes, timetables, vehicles and service requirements are modelled, and how different objectives and constraints shape the resulting transport system.

Finance

Eden Castaneira

Introduces portfolio decision-making under changing estimates of return and risk. Examines how investment mandates, diversification, covariance and rebalancing constraints shape portfolio choices, and why a mathematically optimal solution may change when the underlying assumptions change.

Life Sciences

Prof. Zacharias

Introduces early-stage drug discovery and the role of computational methods in narrowing large sets of molecular candidates. Examines ligands, targets, molecular docking and scoring, and how docking results support decisions about which candidates move forward.

Each session ends with an **expanded methods menu** for its challenge. **Mobility** adds time-expanded graphs, GTFS, fairness measures and QUBOMaker encodings. **Finance** adds cardinality constraints, estimation error and the Qamomile model-to-backend flow. **Life Sciences** adds candidate similarity, submodular objectives and the Zacharias masterclass content. **GovTech** adds substantive versus procedural law, set cover as the shape of a minimal exemption set, and how to build a typology.

SECTION 06

References & reading

Mandatory · recommended · further reading — by track

All tracks MANDATORY

- **Montanez, A. (2026).** Quadratic unconstrained binary optimization. PennyLane Demos. pennylane.ai/demos/tutorial_QUBO
- **Ceroni, J. (2026).** Intro to QAOA. PennyLane Demos. pennylane.ai/demos/tutorial_qaoa_intro
- **Huang, W.-H., et al. (2025).** Qamomile: A cross-SDK bridge for quantum optimization. 2025 IEEE QCE (Vol. 2, pp. 516–517). doi.org/10.1109/QCE65121.2025.10423

All tracks RECOMMENDED

- **Farhi, E., Goldstone, J., & Gutmann, S. (2014).** A quantum approximate optimization algorithm [Preprint]. arXiv. arxiv.org/abs/1411.4028
- **Google (n.d.).** CP-SAT solver. OR-Tools. developers.google.com/optimization/cp/cp_solver
- **Goto, H., Tatsumura, K., & Dixon, A. R. (2019).** Combinatorial optimization by simulating adiabatic bifurcations in nonlinear Hamiltonian systems. Science Advances, 5(4), eaav2372. doi.org/10.1126/sciadv.aav2372

Mobility

MANDATORY

- **Rovara, D., Quetschlich, N., & Wille, R. (2026).** A framework to formulate pathfinding problems for quantum computing. ISMVL 2026 (pp. 158–164). IEEE. doi.org/10.1109/ISMVL68998.2026.00036

RECOMMENDED

- **Dantzig, G. B., & Ramser, J. H. (1959).** The truck dispatching problem. Management Science, 6(1), 80–91. doi.org/10.1287/mnsc.6.1.80
- **Wardrop, J. G. (1952).** Some theoretical aspects of road traffic research. Proc. Institution of Civil Engineers, 1(3), 325–362. doi.org/10.1680/ipeds.1952.11259
- **Stadt Zürich, Open Data Zürich, & VBZ (2025).** Fahrgastzahlen VBZ [Data set]. data.stadt-zuerich.ch
- **Verkehrsbetriebe Zürich (n.d.).** OGD examples Python [Software]. GitHub. github.com/VerkehrsbetriebeZuerich

FURTHER READING

- **Li, Y., Yu, R., Shahabi, C., & Liu, Y. (2018).** Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. ICLR 2018.

Finance

MANDATORY

- **Markowitz, H. (1952).** Portfolio selection. The Journal of Finance, 7(1), 77–91. doi.org/10.1111/j.1540-6261.1952.tb01525.x
- **Qiskit Finance (n.d.).** Portfolio optimization [Tutorial]. qiskit-community.github.io/qiskit-finance
- **French, K. R. (n.d.).** Data library. Tuck School of Business, Dartmouth. mba.tuck.dartmouth.edu

RECOMMENDED

- **Ledoit, O., & Wolf, M. (2004).** Honey, I shrunk the sample covariance matrix. The Journal of Portfolio Management, 30(4), 110–119. doi.org/10.3905/jpm.2004.110
- **Herman, D., et al. (2023).** Quantum computing for finance. Nature Reviews Physics, 5, 450–465. doi.org/10.1038/s42254-023-00603-1
- **Stopfer, E., & Wagner, F. (2025).** Quantum portfolio optimization: An extensive benchmark [Preprint]. arXiv. arxiv.org/abs/2509.17876
- **Sakuler, W., et al. (2025).** A real-world test of portfolio optimization with quantum annealing. Quantum Machine Intelligence, 7, Article 43. doi.org/10.1007/s42484-025-00268-2

FURTHER READING

- **Egger, D. J., et al. (2020).** Quantum computing for finance: State-of-the-art and future prospects. IEEE Transactions on Quantum Engineering, 1, 1–24. doi.org/10.1109/TQE.2020.3030314
- **Orús, R., Mugel, S., & Lizaso, E. (2019).** Quantum computing for finance: Overview and prospects. Reviews in Physics, 4, 100028. doi.org/10.1016/j.revip.2019.100028
- **Acharya, A., et al. (2025).** Decomposition pipeline for large-scale portfolio optimization with applications to near-term quantum computing. Physical Review Research, 7(2), 023142. doi.org/10.1103/PhysRevResearch.7.023142
- **Soloviev, V. P., & Krompiec, M. (2026).** Large-scale portfolio optimization using Pauli correlation encoding. Scientific Reports, 16, 25158. doi.org/10.1038/s41598-026-54244-2
- **Chang, Y. J., et al. (2026).** Quantum stochastic walks for portfolio optimization. npj Unconventional Computing, 3, Article 7. doi.org/10.1038/s44335-025-00050-4

OTHER QUANTUM-FINANCE · NOT PART OF THE CHALLENGE

- **Rebentrost, P., Gupt, B., & Bromley, T. R. (2018).** Quantum computational finance: Monte Carlo pricing of financial derivatives. Physical Review A, 98(2), 022321. doi.org/10.1103/PhysRevA.98.022321
- **Stamatopoulos, N., et al. (2020).** Option pricing using quantum computers. Quantum, 4, 291. doi.org/10.22331/q-2020-07-06-291
- **Miyamoto, K. (2022).** Bermudan option pricing by quantum amplitude estimation and Chebyshev interpolation. EPJ Quantum Technology, 9, Article 3. doi.org/10.1140/epjqt/s40507-022-00124-3
- **Woerner, S., & Egger, D. J. (2019).** Quantum risk analysis. npj Quantum Information, 5, Article 15. doi.org/10.1038/s41534-019-0130-6

Life Sciences

RECOMMENDED

- **Banchi, L., et al. (2020).** Molecular docking with Gaussian Boson Sampling. Science Advances, 6(23), eaax1950. doi.org/10.1126/sciadv.aax1950

FURTHER READING · SEPARATE ELECTRONIC-STRUCTURE TRACK

- **Delgado Gran, A. (2026).** A brief overview of VQE. PennyLane Demos. pennylane.ai/demos/tutorial_vqe
- **Delgado Gran, A. (2026).** Building molecular Hamiltonians. PennyLane Demos. pennylane.ai/demos/tutorial_quantum_chemistry

GovTech

MANDATORY

- **Bundesregierung (2026).** Entwurf eines Gesetzes zur Erprobung neuer Technologien (Bundeserprobungsgesetz, BErpG) [Gesetzentwurf].
- **Bundesministerium für Wirtschaft und Energie (n.d.).** Reallabore-Innovationsportal.
- **Digi-Sandbox.NRW (n.d.).** Register der Reallabore in Nordrhein-Westfalen.

RECOMMENDED

- **Rovara, Quetschlich & Wille (2026)** — see Mobility. The legal reviewer adds one or two readings on substantive versus procedural law.

Note on this document: the visual design and layout were generated with AI. All course content — structure, schedule, readings and descriptions — was authored by the teaching team.